



ASSESSMENT COVERSHEET

Attach this coversheet as the cover of your submission. All sections must be completed.

Section A: Submission Details

Programme : Bachelor of Information Technology in Internet of Things
Course Code & Name : IIB43203 Cloud Computing
Course Lecturer(s) : AP Dr Megat NorulAzmi
Submission Title : Group Project
Deadline : Day 7 Month June Year 2026 Time 11.59pm
Penalties :

- 5% will be deducted per day to a maximum of four (4) working days, after which the submission will **not** be accepted.
- Plagiarised work is an Academic Offence in University Rules & Regulations and will be penalised accordingly.

Section B: Academic Integrity

Tick (✓) each box below if you agree:

☒	I have read and understood the UniKL's policy on Plagiarism in University Rules & Regulations.
☒	This submission is my own, unless indicated with proper referencing.
☒	This submission has not been previously submitted or published.
☒	This submission follows the requirements stated in the course.

Section C: Submission Receipt

(must be filled in manually)

Office Receipt of Submission

Date & Time of Submission (stamp)	Student Name(s)	Student ID(s)
	SYAHMI BIN MUHAMMAD SHAMSUL HASHIR MALIK AHMAD AIDIL IDHAM BIN ZAMRI	52224224168 52227124933 52224224088

Student Receipt of Submission

This is your submission receipt, the only accepted evidence that you have submitted your work. After this is stamped by the appointed staff & filled in, cut along the dotted lines above & retain this for your record.

Date & Time of Submission (stamp)	Course Code	Submission Title	Student ID(s) & Signature(s)
	IIB43203	Create a Web Application using AWS services	52224224168 52227124933 52224224088



**Universiti Kuala Lumpur
Malaysian Institute of Information Technology**

Title:

Create a Web Application using AWS Services

Course:

IIB43203 Cloud Computing

Semester:

March 2026

Prepared by:

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Prepared for:

AP Dr Megat NorulAzmi

1. Introduction

This project was developed to demonstrate how cloud computing services can be used to build and deploy a modern web application. By leveraging Amazon Web Services (AWS), we aimed to integrate Compute, Storage, Networking, and Database solutions into a single functional system. AWS was chosen because it is the leading cloud platform, offering scalability, reliability, and a wide range of services that are accessible through its free trial account. The objective of this project is not only to create a working application but also to provide a comprehensive instructional guide that allows others to replicate the process.

The scope of this project covers the design, deployment, and documentation of a web application using four core AWS service categories:

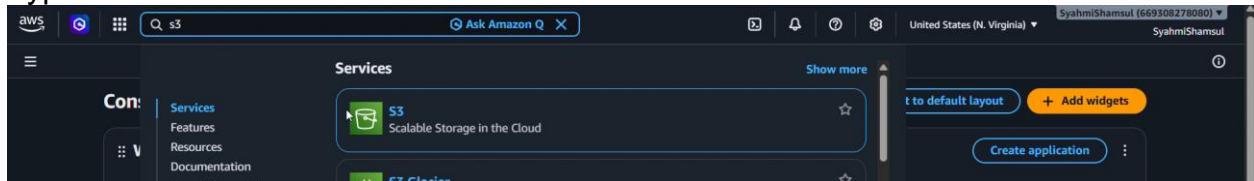
- **Storage (Amazon S3):** Manages and stores files, media, and backups in a secure and scalable environment.
- **Networking (Amazon VPC, Security Groups):** Ensures secure communication between services and controls how data flows in and out of the application.
- **Compute (Amazon EC2):** Provides the processing power to run the application and host the web server.
- **Database (Amazon RDS):** Organizes and manages structured data, enabling efficient storage and retrieval for the application.

This project also includes a step-by-step tutorial guide, a video presentation where all team members contribute, and a GitHub repository containing all project materials. Together, these deliverables showcase teamwork, creativity, and practical cloud computing skills.

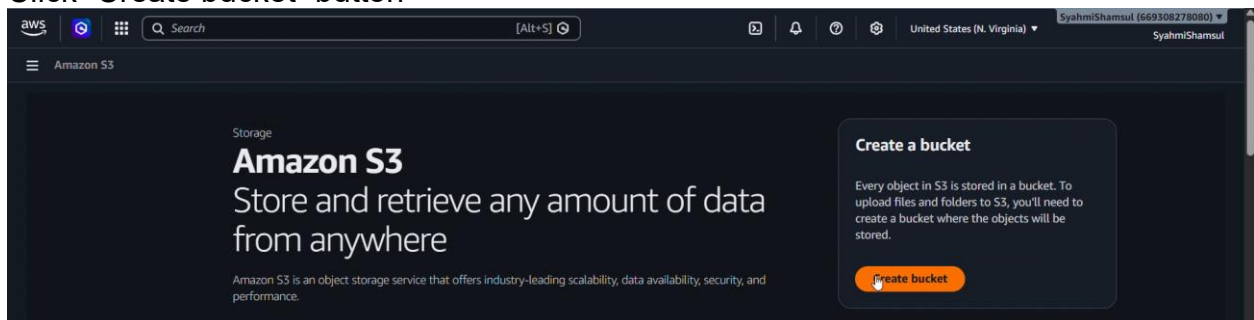
2. Storage (Amazon S3)

Open your AWS console.

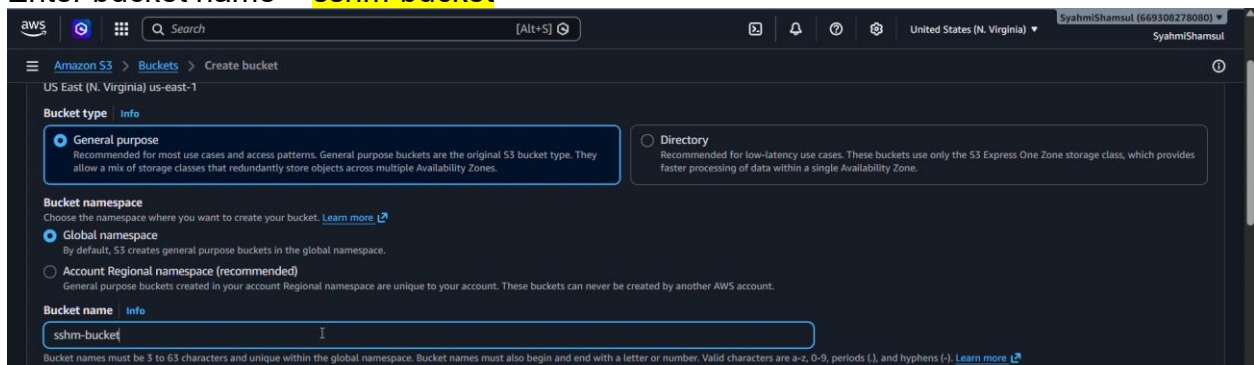
Type “S3” in the search bar.



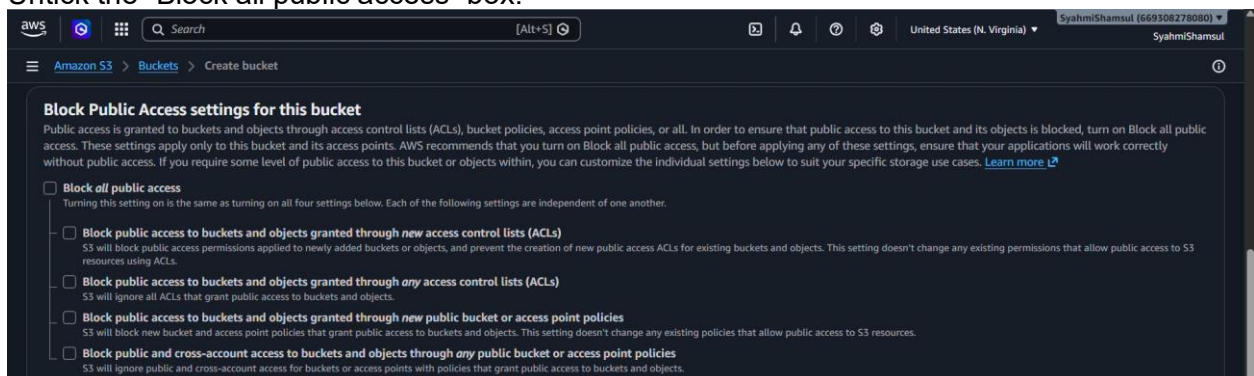
Click “Create bucket” button



Enter bucket name = **sshm-bucket**

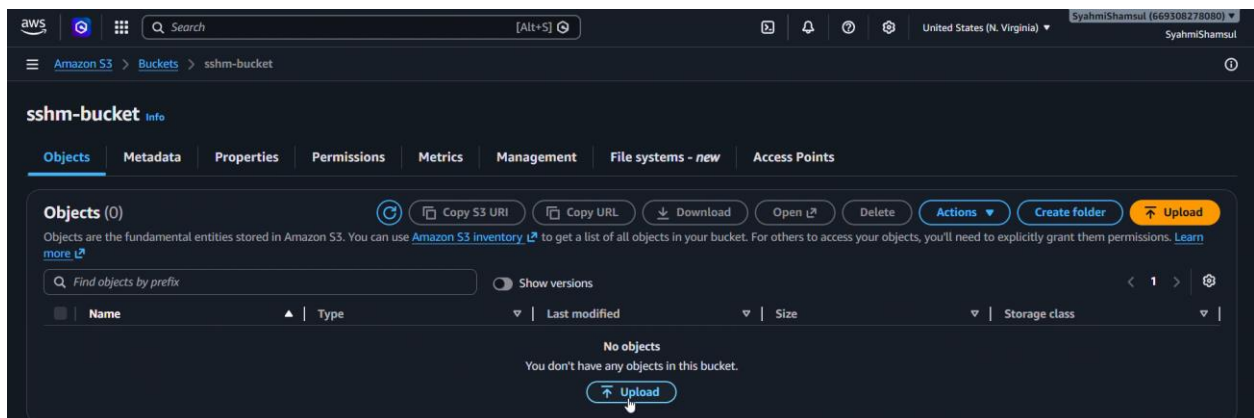


Untick the “Block all public access” box.

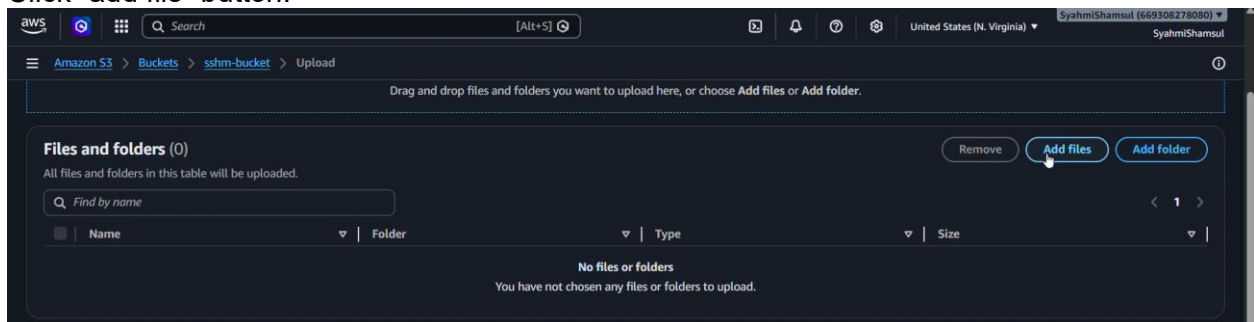


After finished setting up, click the “Create bucket” button at the bottom of the page.

After bucket has been created, click “Upload” button.

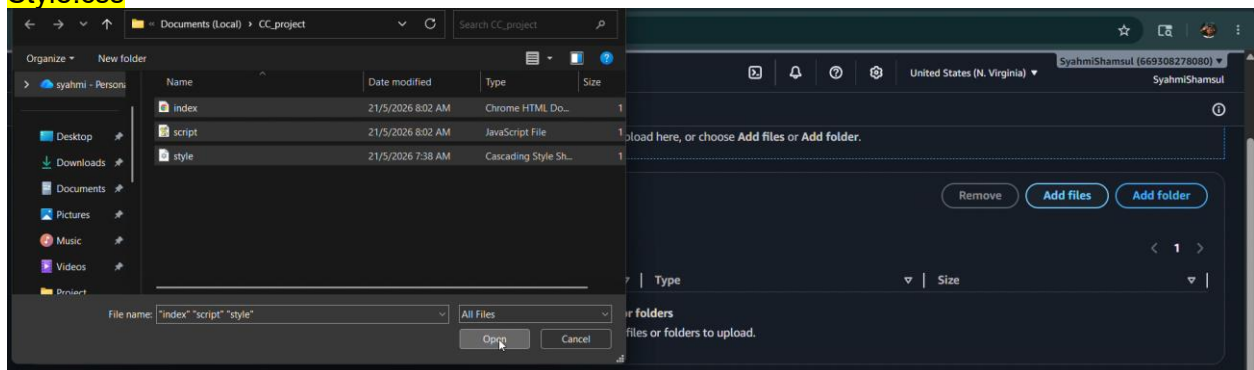


Click “add file” button.



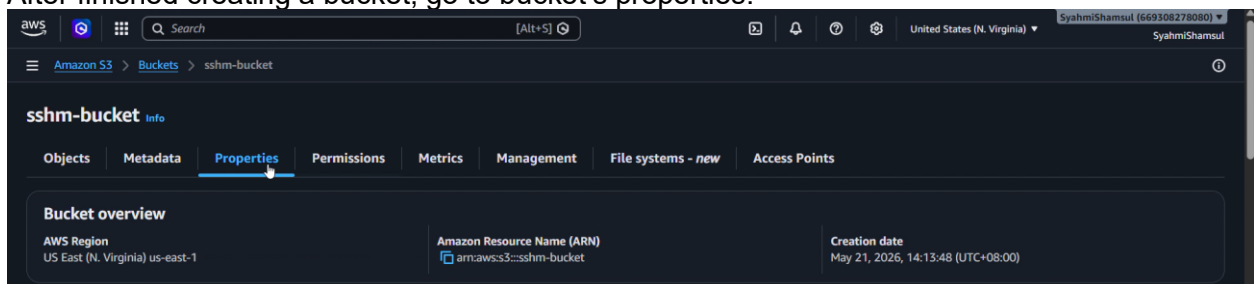
Add files from your computer.

Index.html
Script.js
Style.css

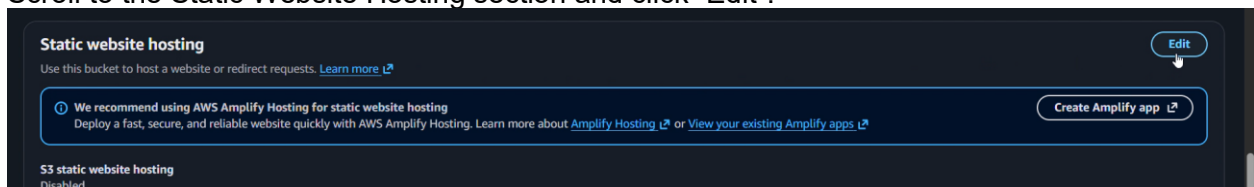


Click “Upload” button at the bottom of the page.

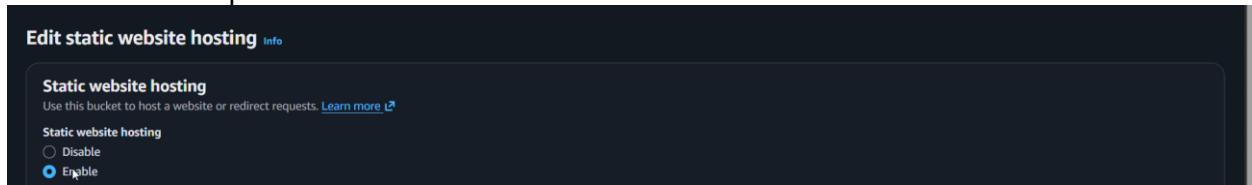
After finished creating a bucket, go to bucket’s properties.



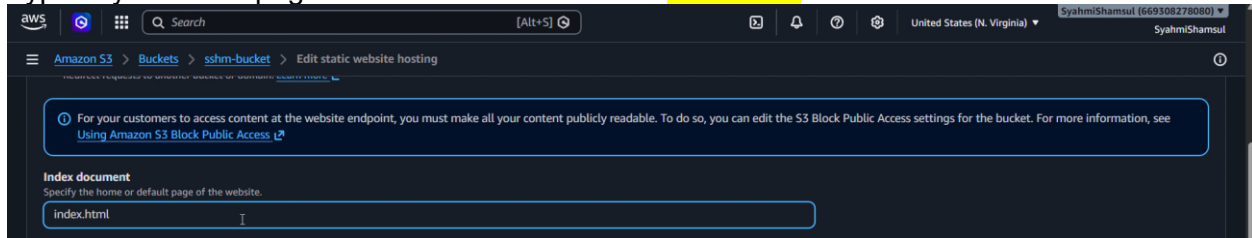
Scroll to the Static Website Hosting section and click “Edit”.



Select “enable” option

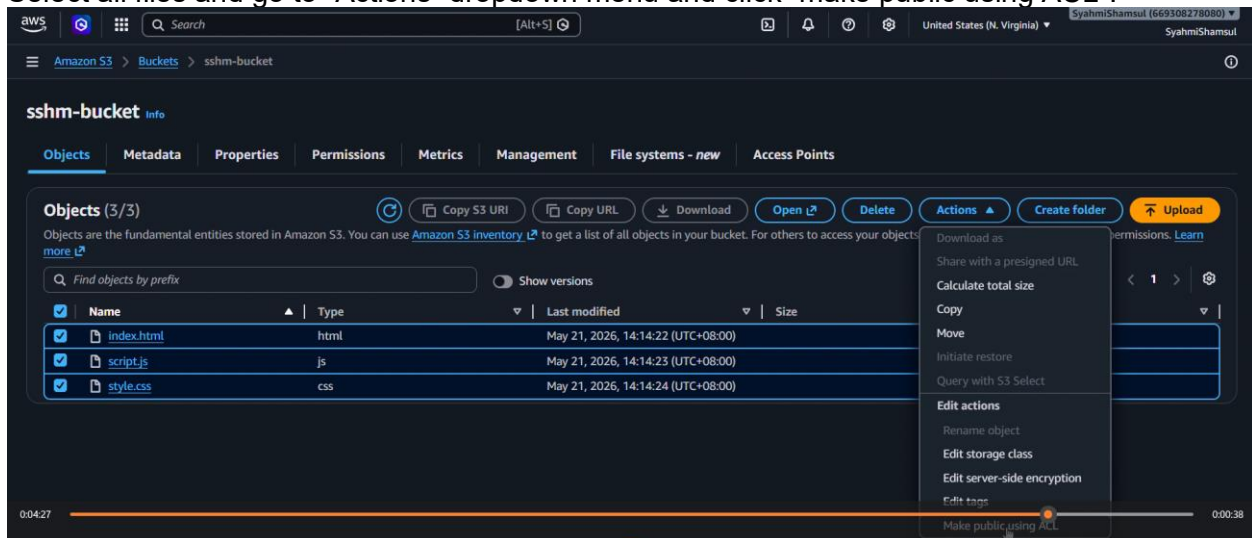


Type in your Homepage files in Index Document = **index.html**

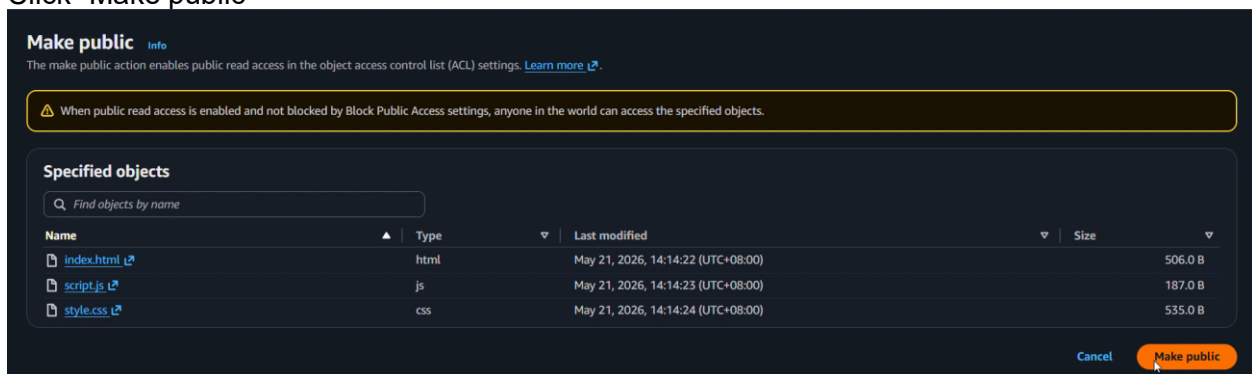


After finished setting up, click “Save changes” button at the bottom of the page. Take note of the website address.

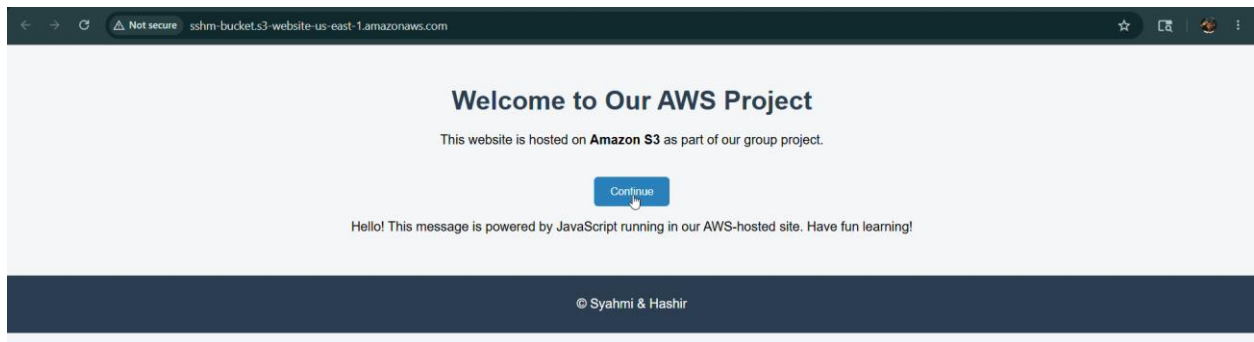
Select all files and go to “Actions” dropdown menu and click “make public using ACL”.



Click “Make public”

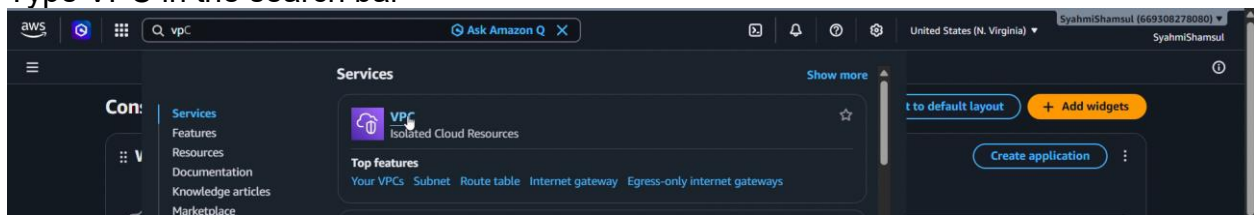


Open the website address



3. Networking (Amazon VPC)

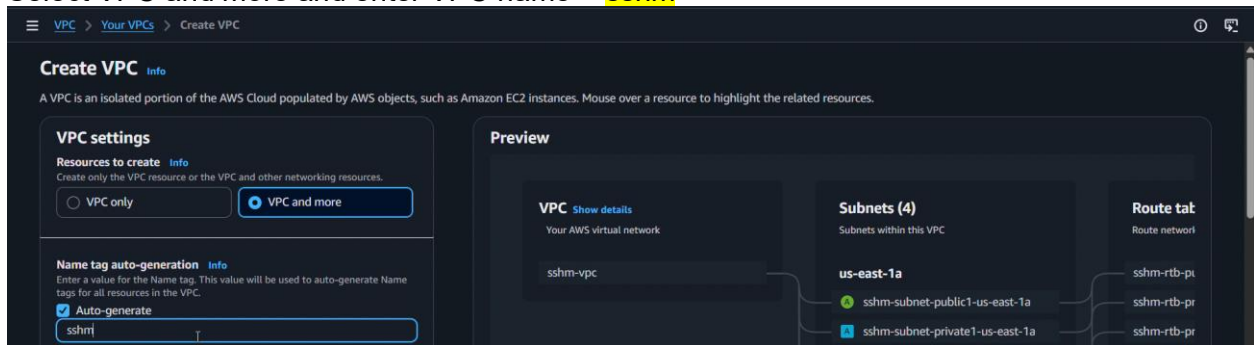
Type VPC in the search bar



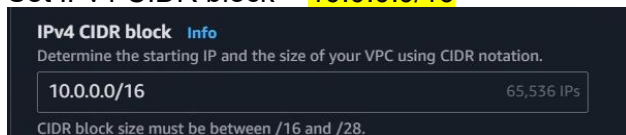
Click "Create VPC" Button



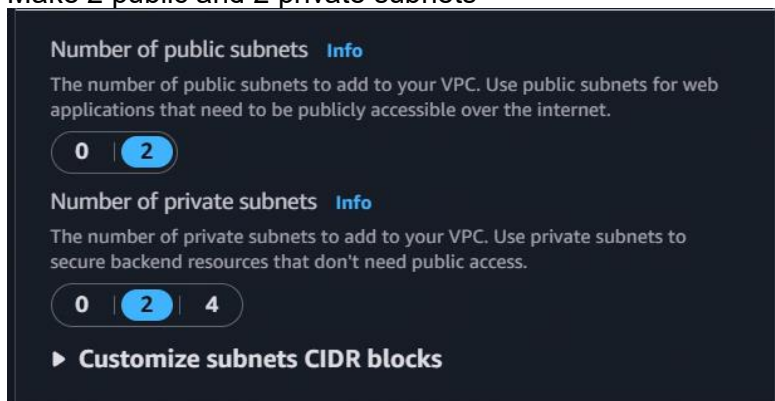
Select VPC and more and enter VPC name = **sshm**



Set IPv4 CIDR block = **10.0.0.0/16**



Make 2 public and 2 private subnets



Inside Customize CIDR block menu, set each subnet CIDR block:

Public-1a = 10.0.0.0/24

Public-1b = 10.0.1.0/24

Private-1a = 10.0.2.0/24

Private-1b = 10.0.3.0/24

▼ Customize subnets CIDR blocks

Public subnet CIDR block in us-east-1a

10.0.0.0/24 256 IPs

Public subnet CIDR block in us-east-1b

10.0.1.0/24 256 IPs

Private subnet CIDR block in us-east-1a

10.0.2.0/24 256 IPs

Private subnet CIDR block in us-east-1b

10.0.3.0/24 256 IPs

< > ^ v

After finished setting up, click “Create VPC” button at the bottom of the page.

Click “Security groups” in lefthand-side of the menu

Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

▼ Security
Network ACLs
Security groups

Your VPCs

VPCs VPC encryption controls

Your VPCs (1/2) Info

Find VPCs by attribute or tag

Last updated 1 minute ago Actions Create VPC

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 C...
sshm-vpc	vpc-0f4a73507f8269bf2	Available	-	-	Off	10.0.0.0/24
-	vpc-0aa89813d31c73864	Available	-	-	Off	172.31.0.0/24

vpc-0f4a73507f8269bf2 / sshm-vpc

Click “Create security group” button

Security Groups (2) Info

Find security groups by attribute or tag

Actions Export security groups to CSV Create security group

Name	Security group ID	Security group name	VPC ID	Description
-	sg-027d53664932d4400	default	vpc-0f4a73507f8269bf2	default VPC security
-	sg-008b83432292c04b5	default	vpc-0aa89813d31c73864	default VPC security

Enter security group name = SSHM-SG

Enter security description

Choose our VPC = sshm-vpc

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

SSHM-SG

Name cannot be edited after creation.

Description Info

Allows access to web

VPC Info

vpc-0f4a73507f8269bf2 (sshm-vpc)

In Inbound rules section, add rule of HTTP, HTTPS and SSH with source of Anywhere-IPv4 (0.0.0.0/0).

Inbound rules info

Type	Protocol	Port range	Source	Description - optional	
HTTP	TCP	80	Anywh... 0.0.0.0		Delete
HTTPS	TCP	443	Anywh... 0.0.0.0		Delete
SSH	TCP	22	Anywh... 0.0.0.0		Delete

[Add rule](#)

After finished setting up, click “Create security group” button at the bottom of the page.

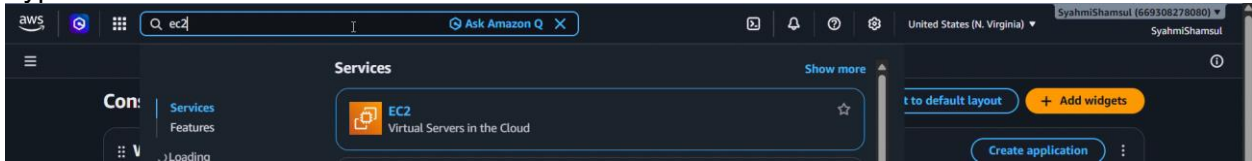
Verify the security group successfully created.

Security Groups (1/3) info [Actions](#) [Export security groups to CSV](#) [Create security group](#)

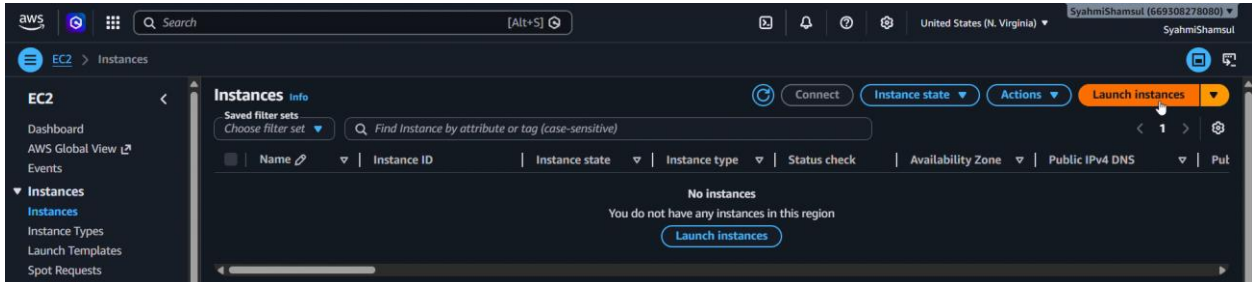
	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-027d53664932d4400	default	vpc-0f4a73507f8269bf2	default VPC security
☐	-	sg-008b83432292c04b5	default	vpc-0aa89813d31c73864	default VPC security
☒	-	sg-020c0f5aba3bf12ab	SSHM-SG	vpc-0f4a73507f8269bf2	SSHM project access

4. Computing (Amazon EC2)

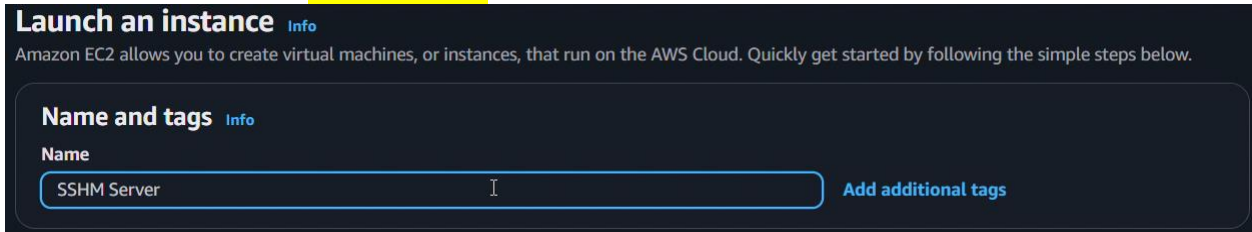
Type in EC2 in the search bar



Click "Launch instances" button



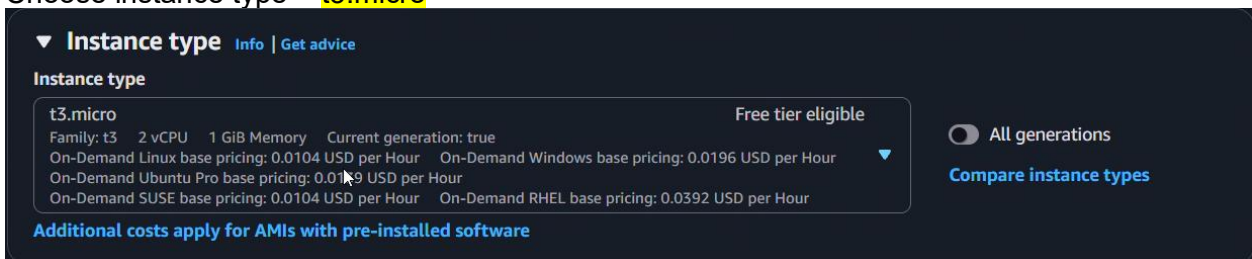
Enter instance name = **SSHM Server**



Choose OS Image = **Ubuntu**



Choose instance type = **t3.micro**



Create new key pair for login process into the instance = **sshm.pem**

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select ▼ [Create new key pair](#)

Create key pair ✕

Key pair name
Key pairs allow you to connect to your instance securely.

sshm

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ **RSA**
RSA encrypted private and public key pair

☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format

☒ **.pem**
For use with OpenSSH

☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

In Network settings section, select your VPC = **sshm-vpc** and any public subnet = **1a**

▼ **Network settings** [Info](#)

VPC - required [Info](#)

vpc-0f4a73507f8269bf2 (sshm-vpc)
10.0.0.0/16

[Create new VPC](#)

Subnet [Info](#)

subnet-0d0782534aa67537c sshm-subnet-public1-us-east-1a
VPC: vpc-0f4a73507f8269bf2 Owner: 669308278080 Availability Zone: us-east-1a (use1-az1)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.0.0/24

[Create new subnet](#)

Enable Auto-assign public IP and select our existing security group = **SSH-M-SG**

Auto-assign public IP [Info](#)

Enable ▼

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#)

Select security groups ▼ [Compare security group rules](#)

SSH-M-SG sg-020c0f5aba3bf12ab ✕
VPC: vpc-0f4a73507f8269bf2

Security groups that you add or remove here will be added to or removed from all your network interfaces.

In the Summary section at the righthand-side, click “Launch Instance” button.

▼ Summary

Number of instances

Info

1

Software Image (AMI)

Canonical, Ubuntu, 26.04...[read more](#)

ami-091138d0f0d41ff90

Virtual server type (instance type)

t3.micro

Firewall (security group)

SSH-M-SG

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Verify the Instance state = **running** and Status check = **3/3 checks passed**

Instances (1/1) Info

Last updated less than a minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

☑

Name

▼

Instance ID

▼

Instance state

▼

Instance type

▼

Status check

▼

Availability Zone

▼

Public IPv4 DNS

▼

Put

☑	SSH-M Server	i-0561482a27a4c80e1	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-3-238-54-43.comp...	3.2
---	--------------	---------------------	---------	----------	-------------------	------------	-------------------------	-----

i-0561482a27a4c80e1 (SSH-M Server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance summary Info

Instance ID

i-0561482a27a4c80e1

Public IPv4 address

3.238.54.43 | [open address](#)

Private IPv4 addresses

10.0.0.220

IPv6 address

-

Instance state

Running

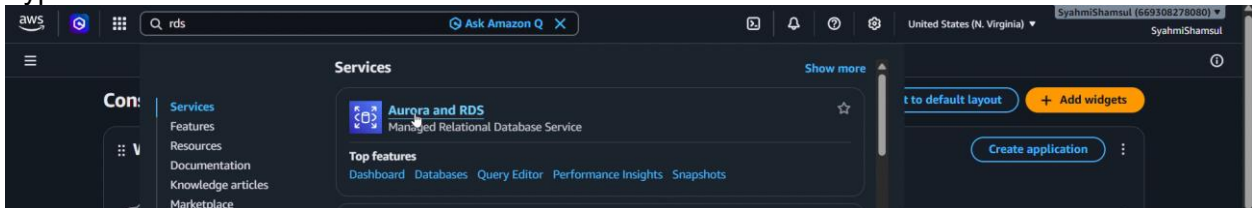
Public DNS

ec2-3-238-54-43.compute-1.amazonaws.com | [open](#)

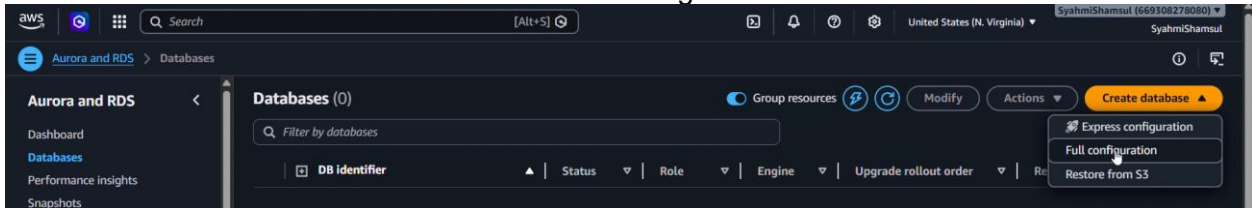
UniKL CDDH v3 Appendix N - Assessment Coversheet v2 (2019-09-17)

5. Database (Amazon RDS)

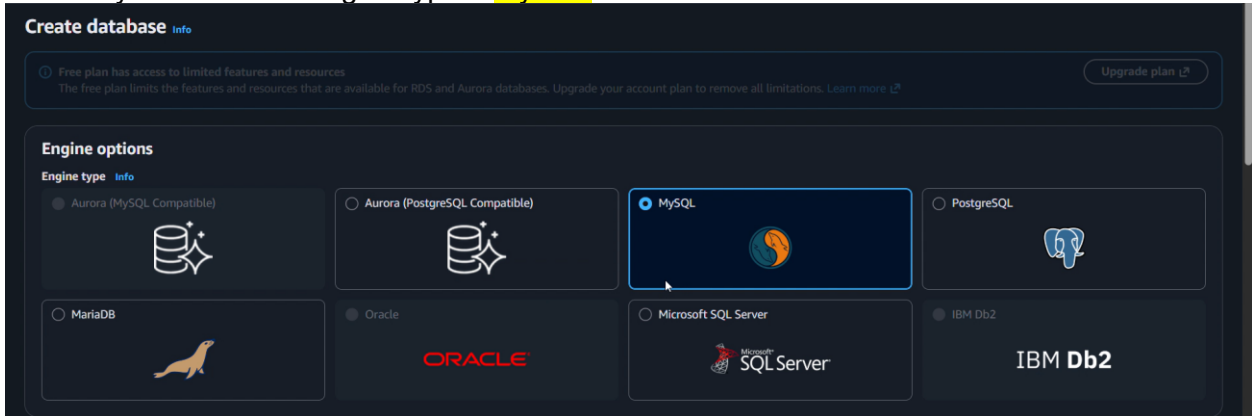
Type RDS in the search bar



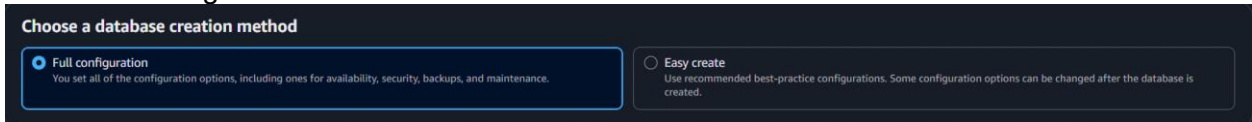
Click "Create database" button and choose full configuration



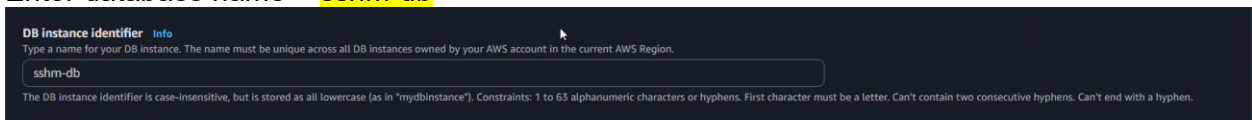
Choose your database engine type = MySQL



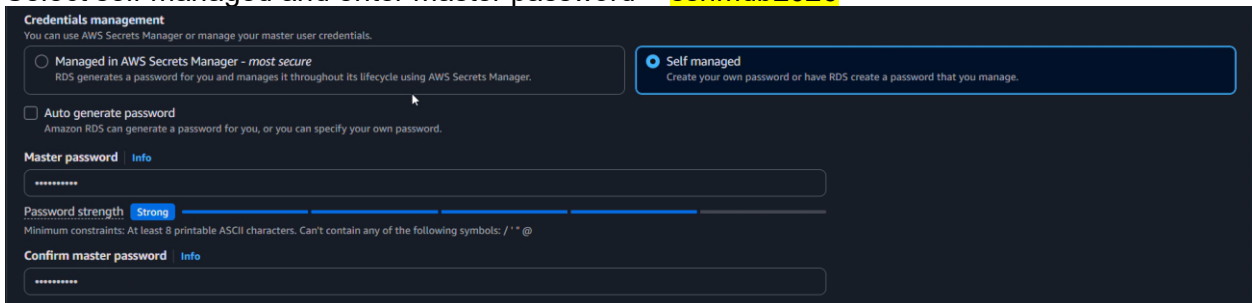
Select full configuration in creation method section



Enter database name = sshm-db



Select self-managed and enter master password = sshmdb2026



Set instance type = t3.micro

Instance type

db.t4g.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Q |

db.t3.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

db.t3.small

In connectivity section, select connect to EC2 compute resource. Then select our EC2 instance = **SSHM Server**

Compute resource
Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☐ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☒ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

EC2 instance Info
Choose the EC2 instance to add as the compute resource for this database. A VPC security group is added to this EC2 instance. A VPC security group is also added to the database with an inbound rule that allows the EC2 instance to access the database.

i-0561482a27a4c80e1
SSHM Server

Virtual private cloud (VPC) Info
Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

sshm-vpc (vpc-0f4a73507f8269bf2)
4 Subnets, 2 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

Select existing security group and choose our security group = **SSHM-SG**

VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ Create new
Create new VPC security group

Additional VPC security group
Choose one or more options

SSHM-SG X

After finished setting up, click "Create database" button at the bottom of the page.

Verify database has successfully been created and status = **available**

Databases (1) Group resources Modify Actions Create database

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size
sshm-db	Available	Instance	MySQL Co...	SECOND	us-east-1a	db.t3.micro

6. How to Run Application

Open terminal in local machine

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\System32> s|
```

Prompt `ssh -i <.pem directory> ubuntu@<EC2 public IP>` to login into your EC2 instance

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\System32> ssh -i C:\Users\cools\Downloads\sshm.pem ubuntu@54.164.204.115|
```

Output:

```
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Thu May 21 09:26:26 UTC 2026

System load:  0.0          Temperature:   ~273.1 C
Usage of /:   35.4% of 6.61GB Processes:    122
Memory usage: 25%         Users logged in: 0
Swap usage:   0%          IPv4 address for ens5: 10.0.0.220

* Ubuntu Pro delivers the most comprehensive open source security and
  compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

30 updates can be applied immediately.
25 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Thu May 21 09:19:29 2026 from 115.135.30.6
ubuntu@ip-10-0-0-220:~$
```

Prompt `sudo apt install apache2 php php-mysql` to install apache inside EC2 instance.

```
ubuntu@ip-10-0-0-220:~$ sudo apt install apache2 php php-mysql
```

Output:

```
apache2 is already the newest version (2.4.66-2ubuntu2.1).
php is already the newest version (2:8.5+99ubuntu1).
php-mysql is already the newest version (2:8.5+99ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 30
```

Prompt `sudo apt install mysql-client` to install mysql client inside EC2 instance.

```
ubuntu@ip-10-0-0-220:~$ sudo apt install mysql-client
```

Output:

```
Installing:
mysql-client

Installing dependencies:
mysql-common

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 30
  Download size: 16.1 kB
  Space needed: 88.1 kB / 4570 MB available

Continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 mysql-common all 5.8+1.1.1ubuntu2 [7002 B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu/resolute/main amd64v3 mysql-client amd64 8.4.8-0ubuntu1 [9136 B]
Fetched 16.1 kB in 0s (826 kB/s)
Selecting previously unselected package mysql-common.
(Reading database ... 86208 files and directories currently installed.)
Preparing to unpack .../mysql-common_5.8+1.1.1ubuntu2_all.deb ...
Unpacking mysql-common (5.8+1.1.1ubuntu2) ...
Selecting previously unselected package mysql-client.
Preparing to unpack .../mysql-client_8.4.8-0ubuntu1_amd64v3.deb ...
Unpacking mysql-client (8.4.8-0ubuntu1) ...
Setting up mysql-common (5.8+1.1.1ubuntu2) ...
update-alternatives: using /etc/mysql/my.cnf.fallback to provide /etc/mysql/my.cnf (my.cnf) in auto mode
Setting up mysql-client (8.4.8-0ubuntu1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.
```

Prompt `sudo nano /var/www/html/<file name>` to create new file inside EC2 acting as a website ui and database connection.

```
ubuntu@ip-10-0-0-220:~$ sudo nano /var/www/html/index2.php
```

Output:

```
GNU nano 8.7.1 /var/www/html/index2.php *
border-radius: 6px;
cursor: pointer;
transition: background-color 0.3s ease;
}
button:hover {
background-color: #45a049;
}
</style>
</head>
<body>
<h1>SSHM Student Information</h1>
<p>Click continue to see student database</p>

<button onclick="window.location.href='sshm_db.php'">
continue
</button>
</body>
</html>
|

ubuntu@ip-10-0-0-220:~$ sudo nano /var/www/html/sshm_db.php
```

Output:

```
<?php
error_reporting(E_ALL);
ini_set('display_errors', 1);

$servername = "sshm-db.cerwuoiquew3.us-east-1.rds.amazonaws.com";
$username = "admin"; // master user
$password = "sshmdb2026"; // master password
$dbname = "sshm_db"; // database you created

$conn = mysqli_connect($servername, $username, $password, $dbname);

if (!$conn) {
die("Connection failed: " . mysqli_connect_error());
}

// Query the students table
$sql = "SELECT id, name, course, year FROM student";
$result = mysqli_query($conn, $sql);

if ($result) {
echo "<h2>✔ Connected successfully!</h2>";
echo "<table border='1' cellpadding='5'>";
echo "<tr><th>ID</th><th>Name</th><th>Course</th></tr>";
while ($row = mysqli_fetch_assoc($result)) {
echo "<tr><td>".$row['id']. "</td><td>".$row['name']. "</td><td>".$row['course']. "</td></tr>";
}
echo "</table>";
} else {
echo "✖ Query failed: " . mysqli_error($conn);
}

mysqli_close($conn);
?>
```

Prompt `sudo systemctl start apache2` to restart apache

```
ubuntu@ip-10-0-0-220:~$ sudo systemctl start apache2
```

Prompt `sudo systemctl enable apache2` to enable apache

```
ubuntu@ip-10-0-0-220:~$ sudo systemctl enable apache2
```

Output:

```
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
```

Prompt `mysql -h <database end point> -p 3306 -u <master username> -p`

```
ubuntu@ip-10-0-0-220:~$ mysql -h sshm-db.cerwuoiquew3.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
```

Enter database password

Enter password:

Output:

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 122
Server version: 8.4.8 Source distribution

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

Enter MySQL code

```
mysql> CREATE DATABASE sshm_db;
Query OK, 1 row affected (0.05 sec)

mysql> USE sshm_db;
mysql> CREATE TABLE student (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    course VARCHAR(50) NOT NULL,
    year INT NOT NULL
);

INSERT INTO student (name, course, year) VALUES
('Syahmi', 'IoT', 3),
('Hasyir', 'IoT', 2),
('Farah', 'AI', 4);

SELECT * FROM student; Database changed
mysql>
mysql> CREATE TABLE student (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    course VARCHAR(50) NOT NULL,
    year INT NOT NULL
);
Query OK, 0 rows affected (0.05 sec)

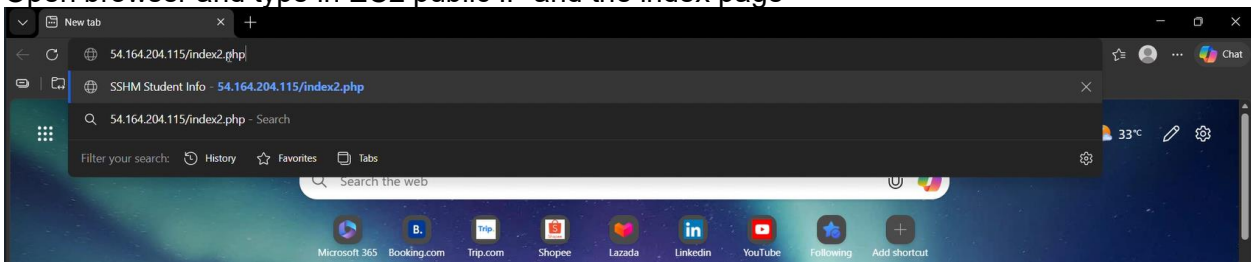
mysql>
mysql> INSERT INTO student (name, course, year) VALUES
    -> ('Syahmi', 'IoT', 3),
    -> ('Hasyir', 'IoT', 2),
    -> ('Farah', 'AI', 4);
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql>
```

Output:

```
mysql> SELECT * FROM student;
+----+-----+-----+-----+
| id | name  | course | year |
+----+-----+-----+-----+
| 1  | Syahmi | IoT    | 3    |
| 2  | Hasyir | IoT    | 2    |
| 3  | Farah  | AI     | 4    |
+----+-----+-----+-----+
3 rows in set (0.00 sec)
```

Open browser and type in EC2 public IP and the index page



Output:

SSHIM Student Info

54.164.204.115/index2.php

Not secure 54.164.204.115/index2.php

SSHIM Student Information

Click continue to see student database

continue

54.164.204.115/sshm_db.php

Not secure 54.164.204.115/sshm_db.php

✓

Connected successfully!

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ID	Name	Course
1	Syahmi	IoT
2	Hasyir	IoT
3	Farah	AI

7. Conclusion

In this project, we successfully designed and deployed a web application using four core AWS services: Compute, Storage, Networking, and Database. Each service played a critical role in ensuring that the application was functional, secure, and scalable. Compute allowed us to run the application on virtual servers, Storage provided a reliable way to manage files and data, Networking ensured secure communication between components, and Database organized and retrieved structured information efficiently.

Through this project, we gained valuable hands-on experience with AWS, learned how different services integrate to form a complete cloud solution, and strengthened our teamwork skills by dividing responsibilities among group members. The deliverables — including the step-by-step guide, video presentation, and GitHub repository — demonstrate not only our technical understanding but also our ability to communicate and document our work clearly.

Overall, this project highlights the importance of cloud computing in modern application development and shows how AWS empowers users to build reliable, scalable, and innovative solutions.

GitHub repository:

https://github.com/MikazeMimi/CloudComputing_GroupProject_March2026.git

Youtube link:

<https://youtu.be/M3D0kO5sATQ>